Yuriy Pichugin

CV

Research interests

Main: Emergence of complexity and self-organization in evolving systems.

evolution of life cycles, major evolutionary transitions, evolutionary game theory, eco-evolutionary dynamics, dynamics of migration.

Research experience

2024 - now **Project leader**, *Max Planck Institute for Evolutionary Biology, Theoretical Biology Department*, Plön, Germany.

2021 - 2024 **Postdoctoral research associate**, *Princeton University, Ecology and Evolutionary Biology department*, Princeton, USA.
Supervisor: Prof. Dr. Corina Tarnita

2016 - 2021 **Postdoctoral researcher**, *Max Planck Institute for Evolutionary Biology, Evolutionary Theory Department*, Plön, Germany.
Supervisor: Prof. Dr. Arne Traulsen

2011 - 2015 **PhD student**, *Massey University, Rainey lab*, Auckland, New Zealand.
Thesis title: Theoretical investigation into the origins of multicellularity. Supervisor: Prof. Dr. Paul B. Rainey

Education

2011–2015 **PhD in Mathematical Biology**, *Massey University*, Auckland, New Zealand.

2008–2010 **Master in Physics**, *Novosibirsk State University*, Novosibirsk, Russia.

2004–2008 **Bachelor in Physics**, *Novosibirsk State University*, Novosibirsk, Russia.

Grants and Awards

2020 Marie Curie Individual Fellowship, EU (grant respectfully declined to accept position at Princeton), EUR 150’000

2015 DAAD short-term research grant, Germany, EUR 9’000

2012 Sir Neil Waters scholarship, New Zealand, NZD 5’000

2010 Schlumberger scholarship, Russia, RUB 50’000

Teaching and mentoring experience

2026 **Project supervisor of the visiting PhD student**, Raiki Nakano, SOKENDAI, Japan.
Project title: Evolution of multicellularity under temporal indirect benefits.

2017 - 2021 **Co-supervisor of the PhD student**, Yuanxiao Gao, University of Luebeck, Germany.
Thesis title: Evolution of life cycles of heterogeneous cell groups.

Apr. - Oct. 2019 **Co-supervisor of the master student**, Vanessa Ress, University of Luebeck, Germany.
Thesis title: Eco-evolutionary dynamics of simple life cycles.

2018 - 2019 **Rotating lecturer in the peer study group "Machine learning"**, Ploen, Germany.

2017 - 2018 **Rotating lecturer in the peer study group "Statistical physics"**, Ploen, Germany.

2008 - 2010 **Teacher of physics**, Specialized Educational Scientific Center (high school equivalent), Novosibirsk, Russia.

Aug. 2006, Aug. 2007, Aug. 2008 **Teacher of physics**, Summer school at SESC, Novosibirsk, Russia.

Community service and organizational experience

Guest editor, PLOS Computational Biology.

Reviewer, for: BMC Biology, BMC Ecology and Evolution, Dynamic Games and Applications, Ecology Letters, Journal of Theoretical Biology, Nature Biomedical Engineering, Nature Communications, PeerJ, PLOS Computational Biology, PLOS One, Proceedings B.

Sep. 2019 **Co-organiser of the workshop "Evolution of interacting populations"**, Max Planck Institute for Evolutionary Biology, Ploen, Germany.

Jul. 2019 **Organiser of the mini-symposium "Evolution of multicellular life cycles"**, part of the international conference MMEE-2019, Lyon, France.

2018 - 2019 **Initiator and coordinator of the peer study group "Machine learning"**, Max Planck Institute for Evolutionary Biology, Ploen, Germany.

Publication list

∗ indicates shared first authorship
✉ indicates corresponding authorship

2026 D. Jorge*, M.Staps*, **Y.Pichugin***, C.Tarnita, Direct benefits are not necessary for the evolution of multicellularity, *Nature Ecology and Evolution*, 1-12.

2025 Y. Gao, **Y.Pichugin**, A. Traulsen, R. Zapién Campos, Evolution of irreversible differentiation under stage-dependent cell differentiation, *Scientific Reports* 15, 7786.

2023 S. Tang*, **Y.Pichugin***✉, K.Hammerschmidt, An environmentally induced multicellular life cycle of a unicellular cyanobacterium, *Current Biology* 13(4), 764-769.e5.

2022 V.Ress, A.Traulsen, **Y.Pichugin**✉, Eco-evolutionary dynamics of clonal multicellular life cycles, *eLife* 11, e78822.

2022 J.T.Lange*, J.C.Rose*, C.Y.Chen*, **Y.Pichugin***, L. Xie, J. Tang, K.L. Hung, K.E.Yost, Q. Shi, M.L. Erb, U. Rajkumar, S. Wu, S. Taschner-Mandl, M. Bernkopf, C. Swanton, Z. Liu, W. Huang, H.Y. Chang, V. Bafna, A.G. Henssen, B. Werner, P.S. Mischel, The evolutionary dynamics of extrachromosomal DNA in human cancers, *Nature Genetics* 54, 1527–1533.

2022 **Y.Pichugin**, A. Traulsen, The possible modes of microbial reproduction are fundamentally restricted by distribution of mass between parent and offspring, *Proceedings of the National Academy of Sciences* 119 (12), e2122197119.

2022 Y.Gao, **Y.Pichugin**, C.S.Gokhale, A. Traulsen, Evolution of reproductive strategies in incipient multicellularity, *Journal of the Royal Society Interface* 19 (188), 20210716.

2021 Y.Gao, H.J.Park, A. Traulsen, **Y.Pichugin**✉, Evolution of irreversible somatic differentiation, *eLife* 10, e66711.

2021 O. Yaryhin, J. Klembara, **Y.Pichugin**, M. Kaucka, I. Werneburg, Limb reduction in squamate reptiles correlates with the reduction of the chondrocranium: a case study on serpentiform anguids, *Developmental Dynamics*, 1-18.

2020 **Y.Pichugin**✉, A.Traulsen, Evolution of multicellular life cycles under costly fragmentation, *PLoS Computational Biology*, 16 (11), e1008406.

2020 H.J. Park, **Y.Pichugin**, A.Traulsen, Why is cyclic dominance so rare?, *eLife* 9, e57857.

2020 C.J. Rose, K. Hammerschmidt, **Y.Pichugin**, P.B.Rainey, Meta-population structure and the evolutionary transition to multicellularity, *Ecology Letters* 23 (9), 1380-1390.

2019 **Y.Pichugin**✉, H.J.Park, A.Traulsen, Evolution of simple multicellular life cycles in dynamic environments, *Journal of the Royal Society Interface* 16 (154), 20190054.

2019 H.J.Park, **Y.Pichugin**, W.Huang, A.Traulsen, Population size changes and extinction risk of populations driven by mutant interactors, *Physical Review E* 99 (2), 022305.

2019 Y.Gao, A.Traulsen, **Y.Pichugin**✉, Interacting cells driving the evolution of multicellular life cycles, *PLoS Computational Biology* 15(5), e1006987.

2017 **Y.Pichugin**✉, J.Peña, P.B.Rainey, A.Traulsen, Fragmentation modes and the evolution of life cycles, *PLoS Computational Biology* 13 (11), e1005860.

2015 **Y.Pichugin**✉, C.S.Gokhale, J.Garcia, A.Traulsen, P.B.Rainey, Modes of migration and multilevel selection in evolutionary multiplayer games, *Journal of Theoretical Biology* 387, 144-153.

2012 **Y.G.Pichugin**, K.A.Semiyanov, A.V.Chernyshev, I.G.Palchikova, L.V.Omelyanchyuk, V.P.Maltsev, Peculiarities of cytometrical methods of DNA content determination in the nucleus, *Cell and Tissue Biology* 6 (3), 302-308.

Invited talks

2026 **Evolution of informed dispersal strategies in trophic meta-communities**, CCTB institute seminar, Würzburg, Germany.

2026 **Evolution of informed dispersal strategies in trophic meta-communities**, The Arctic Meeting for Adaptive Mechanisms in Biological Systems, Abisko, Sweden.

2025 **Evolution of multicellularity: a theoretical framework**, Mathematical modeling seminar, University of Oldenburg, Germany.

2025 **Evolution of informed dispersal strategies in trophic meta-communities**, Modeling in Ecology and Evolution meeting, Incheon, South Korea, *online talk*.

2024 **Impact of informed dispersal on the stability of trophic communities**, Research Training Group TransEvo core seminar, Plön, Germany.

2024 **Evolution of somatic cell differentiation at the onset of multicellularity**, conference Evolving Individuality, Sydney, Australia.

2024 **A theory of multicellularity**, Fellowship selection symposium, UCL, London, England, *online talk*.

2024 **A theory of multicellularity**, FASSBERG-Colloquium, Max Planck Institute for Multidisciplinary Sciences, Göttingen, Germany.

2024 **Evolution of somatic cell differentiation at the onset of multicellularity**, Multicellgenome lab seminar, Institut de Biologia Evolutiva (CSIC-Universitat Pompeu Fabra), Barcelona, Spain, *online talk*.

2023 **A theory of multicellularity**, Research group leader selection symposium, Max Planck Society, Berlin, Germany.

2022 **A theory of multicellularity**, Group leader selection symposium, EMBL, Heidelberg, Germany.

2022 **Eco-evolutionary dynamics of multicellular life cycles**, Modeling on Ecology and Evolution Workshop, KIAS, Seoul, South Korea, *online talk*.

2022 **Mathematics of multicellularity**, University of Toronto, Ecology and Evolutionary Biology department seminar, Toronto, Canada.

2022 **Evolution of irreversible somatic differentiation**, Online seminar series Modelling Cell Development and Regeneration, *online talk*.

2020 **Stochastic dynamics of ecDNA in tumors**, Theory division seminar at Cleveland clinic, Cleveland, USA, *online talk*.

2019 **Evolution of simple multicellular life cycles**, Biomathematics seminar at Harvard University, Boston, USA.

2018 **Evolution of life cycles in early multicellularity**, Complex systems seminar at Queen Mary University of London, UK.

2017 **Fitness correlation metric as a method of detection of the transition in individuality**, Watson group seminar, University of Southampton, UK.

Contributed talks and posters on scientific meetings

2025 **Multicellularity could have evolved even in the absence of direct benefits**, EMBO workshop "Evolution and origins of multicellularity across the tree of life", Barcelona, Spain (*poster*).

2025 **Evolution of irreversible somatic differentiation and the evolution of multicellular life cycles**, Jacques Monod conference "Origin of metazoans", Roscoff, France (*poster*).

2022 **The possible modes of microbial reproduction are fundamentally restricted by distribution of mass between parent and offspring**, AbSciCon-2022, Atlanta, USA (*talk*).

2021 **Mass conservation restricts the possible modes of microbial reproduction**, SMB-2021: Society of Mathematical Biology annual meeting, online conference (*online talk*).

2020 **Evolution of clonal life cycles: recipes for multicellularity, equal split, and single cell bottleneck**, SMB-2020: Society of Mathematical Biology annual meeting, online conference (*online talk*).

2019 **Evolution of simple multicellular life cycles**, Evolution of interacting populations workshop, Ploen, Germany (*talk*).

2019 **Towards a general theory of reproduction modes evolution**, MMEE-2019: Mathematical models in Ecology and Evolution, Lyon, France (*talk*).

2019 **Evolution of life cycles in early multicellularity**, CCCC-2019: Conflict, Competition, Cooperation and Complexity, Vaals, Netherlands (*talk*).

2018 **Evolution of simple multicellular life cycles in a dynamic environment**, Evolutionary emergence of life cycles workshop, Ploen, Germany (*talk*).

2018 **Evolution of life cycles in early multicellularity**, Evolutionary Models of Structured Populations: Integrating Methods workshop, Ploen, Germany (*talk*).

2018 **Reproduction costs can drive the evolution of groups**, ECMTB-2018: European Conference on Mathematical and Theoretical Biology, Lisbon, Portugal (*talk*).

2018 **Evolution of simple multicellular life cycles in a dynamic environment**, MPDEE-2018: Models in Population Dynamics, Ecology, and Evolution, Leicester, UK (*talk*).

2018 **Evolution of simple multicellular life cycles in a dynamic environment**, DPG-2018: Deutsche Physikalische Gesellschaft meeting, Berlin, Germany (*poster*).

2017 **Fragmentation modes and the evolution of multicellular life cycles**, MMEE-2017: Mathematical models in Ecology and Evolution, London, UK (*talk*).

2017 **Fitness correlation metric as a method of detection of the transition in individuality**, MBE-2017: Modelling Biological Evolution, Leicester, UK (*talk*).

2016 **Fragmentation modes and the evolution of multicellular life cycles**, CCCC-2016: Conflict, Competition, Cooperation and Complexity, Prague, Czech Republic, (*talk*).

2016 **Fragmentation strategies and the evolution of multicellular life cycles**, ECMTB-2016: European Conference on Mathematical and Theoretical Biology, Nottingham, UK (*poster*).

2015 **Modes of migration and multilevel selection in evolutionary multiplayer games**, MMEE-2015: Mathematical models in Ecology and Evolution, Paris, France (*poster*).

Non peer-reviewed publications and outreach activities

2020 **Antibody tests and our return to a more normal life**, *Article in the scientific outreach web-site on the COVID pandemic*, Co-author and translator to Russian, `http://web.evolbio.mpg.de/evoltheo_corona/articles/YP_AntibodyTests/index_EN.html`.

2020 **Exponential virus growth and increasing the capacity of a health-care system**, *Article in the scientific outreach web-site on the COVID pandemic*, Co-author, `http://web.evolbio.mpg.de/evoltheo_corona/articles/AT_CommExpGrowth/01_english/Eng_ExponentiellesWachstum_May.pdf`.

2019 **Was können uns mathematische Modelle über die Evolution von einfachen Lebenszyklen sagen?**, *Chapter in Max Planck Society Year book (in German)*, Co-author, Munich, Germany.

2019 **The game of life cycles**, *The demonstration boardgame presented during European Researcher's Night*, Designer and demonstrator, Ploen, Germany.

2018 **From one to many**, *Popular science article*, Author, New Principia. http://www.newprincipia.com, Pilot issue 001 "Time".

2018 **The game of life cycles**, *The demonstration boardgame presented during Max Planck Institute open day*, Designer and demonstrator, Ploen, Germany.

2018 **Die Evolution von Lebenszyklen**, *Popular science article (in German)*, Co-author, EvolBioMax magazine.

2018 **Write your name with DNA**, *The demonstration at European Researcher's Night*, Demonstrator, Kiel, Germany.

2017 **Write your name with DNA**, *The demonstration at Max Planck Institute open day*, Demonstrator, Ploen, Germany.

Contact information

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Google scholar **https://scholar.google.com/citations?hl=en&user=8xGTdC8AAAAJ**